



rm -rf doubts

Trusting an Agent with your production environment

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Agents have become a trend, but realizing value from them is not straightforward

Elite DevOps teams deploy

208x

More frequently with 106x faster lead times than low performer

(source: DORA State of DevOps Report, 2024)

16.2%

Of organisations achieve on-demand deployment, while the majority still deploy weekly or slower

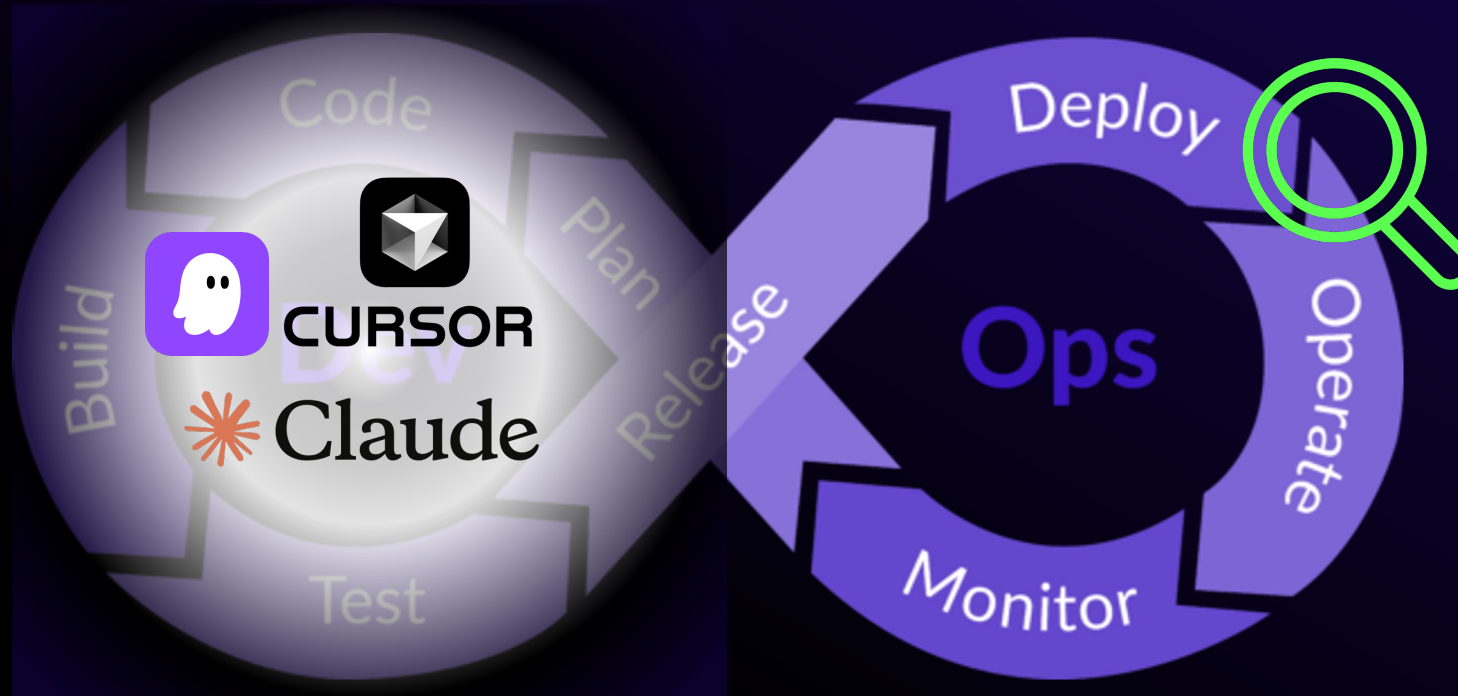
(source: DORA Metrics Benchmark, 2025)

37%

Of IT leaders say DevOps is their team's biggest technical skill

(source: Spacelift, 2026)

As IT infrastructure grows more complex, AI agents are becoming central to modern DevOps



DevOps engineers need scarce, multi-domain expertise to keep digital services reliable, secure, and resilient



Release & Change execution

- Terraform/ Kubernetes config changes
- CI/CD pipeline setup and workflows
- Rollout and rollback strategy



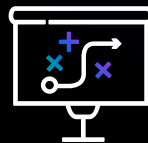
Runtime observation

- Infrastructure: CPU, memory, network
- Applications/DBs: latency, errors, transactions, read/write throughput



Detection and triage

- Anomalies, outages, failed deployments
- Correlate alerts and prioritize severity
- Narrow down affected components



Diagnosis & Decision

- Analyse logs, metrics, traces, and recent changes
- Assess user/business impact
- Decide: rollback, patch, restart, scale, escalate

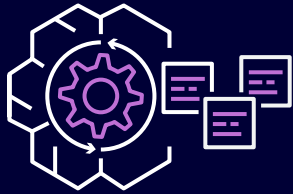


Response & learning

- Execute remediation and communicate status
- Estimate recovery / coordinate incident response
- Run postmortems and improve monitoring, runbacks, and automation

Pairing dynamic tool selection with deterministic safety enforcement for production-grade reliability

Dynamic Decision Making



API selection based on incident context



Log pattern identification of and filtering

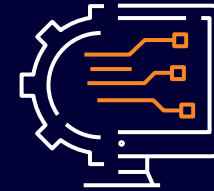


Adaptive depth of signal investigation



Autonomous retrieval of additional evidence

Deterministic Safety Boundaries



Agent Policy (Cedar): enforce read only actions



Automated reasoning to verify conclusions against operational rules



Bedrock Guardrails for PII redaction and content safety



IAM roles to scope API access per account and region

Building end-to-end traceability through immutable logging and stepwise accountability



Immutable Agent Journals for tamper proof decision auditing



Step Functions or deterministic state tracking and error handling



Cloud Trail for continuous API-level monitoring and compliance



DynamoDB persistence at each state for stepwise accountability

Enforcing least privilege through identity-native controls that isolate, scope, and revoke agent access

Separate Agent Spaces isolate environments

Each on-call team operates within its own logical boundary with dedicated IAM roles per account, preventing a staging investigation from accessing production resources

IAM roles with short-lived credentials enforce least privilege per account

Agents assume scoped read-only roles via no long-lived keys, region-restriction and destructive actions explicitly denied

Centralised revocation through identity-layer controls

In case of unexpected agent behaviour, security teams disable access at the IAM role level, immediately cutting off access across accounts

Just-in-time privilege escalation for cross-boundary investigations

When an investigation requires access beyond the Agent Space boundary, temporary scoped credentials can be issued with Cedar policy evaluation and full Cloud Trail audit

Lessons Learned

Understanding the boundaries of what an agent can and cannot resolve is crucial for its application



Detection is not prevention, with agents reacting to symptoms not architectural root causes



Cross-boundary investigations expose the limits of least privilege



Recovery requires business context that infrastructure monitoring cannot provide



The agent's understanding is bounded by its topology

Thank you!

